THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. In science, a model is best described as: 1

(a) a simplified representation of a system that keeps what matters for the question
(b) an exact copy of the real object
(c) an untested guess
(d) a final truth that never changes

2. A pure substance consists of: 1

(a) a single kind of particle with fixed composition
(b) only metals
(c) two or more substances in any ratio
(d) only gases

3. The law of conservation of mass was proposed by: 1

(a) Joseph Proust **(b)** Antoine Lavoisier
(c) J. J. Thomson **(d)** John Dalton

4. The sub-atomic particle discovered by J. J. Thomson was the: 1

(a) nucleus **(b)** electron **(c)** proton **(d)** neutron

5. A runner completes one full lap of a 400 m circular track. Her displacement at the end is: 1

(a) 400 m **(b)** 800 m **(c)** 200 m **(d)** 0

6. A body continues in its state of rest or of uniform motion in a straight line unless acted on by an external force. This is: 1

(a) Newton's first law **(b)** Newton's second law
(c) the law of gravitation **(d)** Newton's third law

7. In science, work is done when: 1

(a) a force acts on a body and the body is displaced in the direction of the force
(b) a force acts but there is no displacement
(c) a person holds a heavy load at rest
(d) a person thinks hard

8. Sound is produced by: 1

(a) vibrating bodies **(b)** heat only
(c) stationary bodies **(d)** light

9. The cell was first observed and named by: 1

(a) Anton van Leeuwenhoek **(b)** Robert Hooke
(c) Robert Brown **(d)** Rudolf Virchow

10. A group of similar cells performing a specific function is called a: 1

(a) organelle **(b)** organism **(c)** tissue **(d)** organ

11. The variety of living organisms found on the Earth is called: 1

(a) taxonomy **(b)** biodiversity **(c)** heredity **(d)** ecology

12. Reproduction in which a single parent produces offspring identical to itself is: 1

(a) fertilisation **(b)** sexual reproduction
(c) pollination **(d)** asexual reproduction

13. The four spheres of the Earth system are the geosphere, hydrosphere, atmosphere and: 1

(a) troposphere **(b)** stratosphere **(c)** biosphere **(d)** lithosphere only

14. A rough estimate is useful when: 1

(a) an exact value is required
(b) we want to avoid reasoning
(c) an approximate answer is enough for a decision
(d) no data are available at all

15. A suspension is: 1

(a) a heterogeneous mixture whose particles settle on standing
(b) a pure substance
(c) a clear solution
(d) a compound

16. The atomicity of phosphorus (P_4) is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 8

17. The atomic number of an element is equal to the number of: 1

(a) neutrons **(b)** nucleons
(c) valence electrons only **(d)** protons

18. The slope of a velocity-time graph gives: 1

(a) distance **(b)** speed **(c)** force **(d)** acceleration

19. According to Newton's third law, action and reaction forces: 1

(a) cancel each other on the same body
(b) are equal in magnitude, opposite in direction and act on different bodies
(c) are unequal
(d) act on the same body

20. The work-energy theorem states that the work done on a body is equal to the change in its: 1

(a) momentum **(b)** potential energy **(c)** kinetic energy **(d)** mass

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. What is a model in science? Give one example from chemistry. 2

22. Differentiate between a homogeneous and a heterogeneous mixture with one example each. 2

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23. State the law of conservation of mass. Who proposed it? 2

24. State two limitations of Thomson's model of the atom. 2

25. Differentiate between distance and displacement. 2

26. State Newton's first law of motion. What is inertia? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Explain with an example how a model is built and why assumptions are made. 3

28. Explain the method of separating a mixture of salt, sand and ammonium chloride. 3

29. A reaction between 6.0 g of carbon and 16.0 g of oxygen gives CO_2. Verify the law of conservation of mass. 3

30. Describe the gold foil experiment: set-up, observations and the conclusion. 3

31. A boy walks 3 m east and then 4 m north. Find the distance and the magnitude of displacement. 3

OR

A body of mass 2 kg is acted on by forces of 8 N east and 3 N west. Find the net force and acceleration. 3

32. A girl lifts a dumbbell of 5 kg to a height of 1.2 m and then lowers it. State when positive and when negative work is done and calculate the work done by her force in lifting (g = 10). 3

OR

A wave has a frequency of 200 Hz and a wavelength of 1.7 m. Find its speed and time period. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Explain the characteristics of the scientific method with an example of an investigation, including question, hypothesis, experiment, observation and conclusion. 5

34. Differentiate between pure substances and mixtures. Classify each as pure, homogeneous mixture or heterogeneous mixture: air, sugar solution, gold, milk, brass. 5

35. Describe an experiment to verify the law of conservation of mass using sodium sulphate and barium chloride solutions in a closed flask. State the observation and conclusion. 5

36. Describe Thomson's, Rutherford's and Bohr's models of the atom with one drawback of each of the first two. 5

OR

Derive the three equations of motion for a body moving with uniform acceleration using the velocity-time graph or algebra. 5

37. State Newton's three laws of motion and give one example of each. 5

OR

Explain the work done by a constant force, with the cases of force along, opposite and perpendicular to the displacement. Calculate the work done by a force of 15 N moving a body 4 m along the force. 5

38. Explain the production and propagation of sound waves, with the definitions of compression, rarefaction, amplitude, wavelength, frequency and time period. 5

OR

Describe the structure of a typical animal cell with a labelled description of the organelles and their functions. 5

END OF QUESTION PAPER **Total: 80 Marks**

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